

MODELING BELUGA WHALE POPULATION DYNAMICS IN COOK INLET

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FW 662

Background

- Geographically isolated
- Genetically distinct
- Abundance decreasing since 1970s
- Protected under ESA in 2008



Abundance Estimation

- NMML has monitored abundance since 1993
- Multiple surveys/year within a ~ 10 day period

Year	Number of individuals counted				
	Survey 1	Survey 2	Survey 3	Survey 4	...
2004	251	177	201	198	
2005	149	189	215	263	
2006	180	213	243	257	
2007	291	193	211	227	
⋮					



Imperfect Detection

- Biases abundance and coefficient estimates
- Use video and multiple observers to correct for bias:
 - Whales not at surface
 - Conditions and camouflage
 - Whales not encountered during survey



$\mathcal{N}$ -mixture Model

$$y_i \sim \text{Binom}(N, p_i) \quad \text{Observation model}$$

$$N \sim \text{Pois}(\theta) \quad \text{Biological process model}$$

$$\text{logit}(\mathbf{p}) = \mathbf{W}\boldsymbol{\alpha} \quad \text{Detection model}$$

$$\theta \sim \text{Gamma}(r, q) \quad \text{Prior distributions}$$

$$\boldsymbol{\alpha} \sim \text{N}(\boldsymbol{\mu}_\alpha, \tau^2 \mathbf{I})$$

y_i — count of individuals during survey i

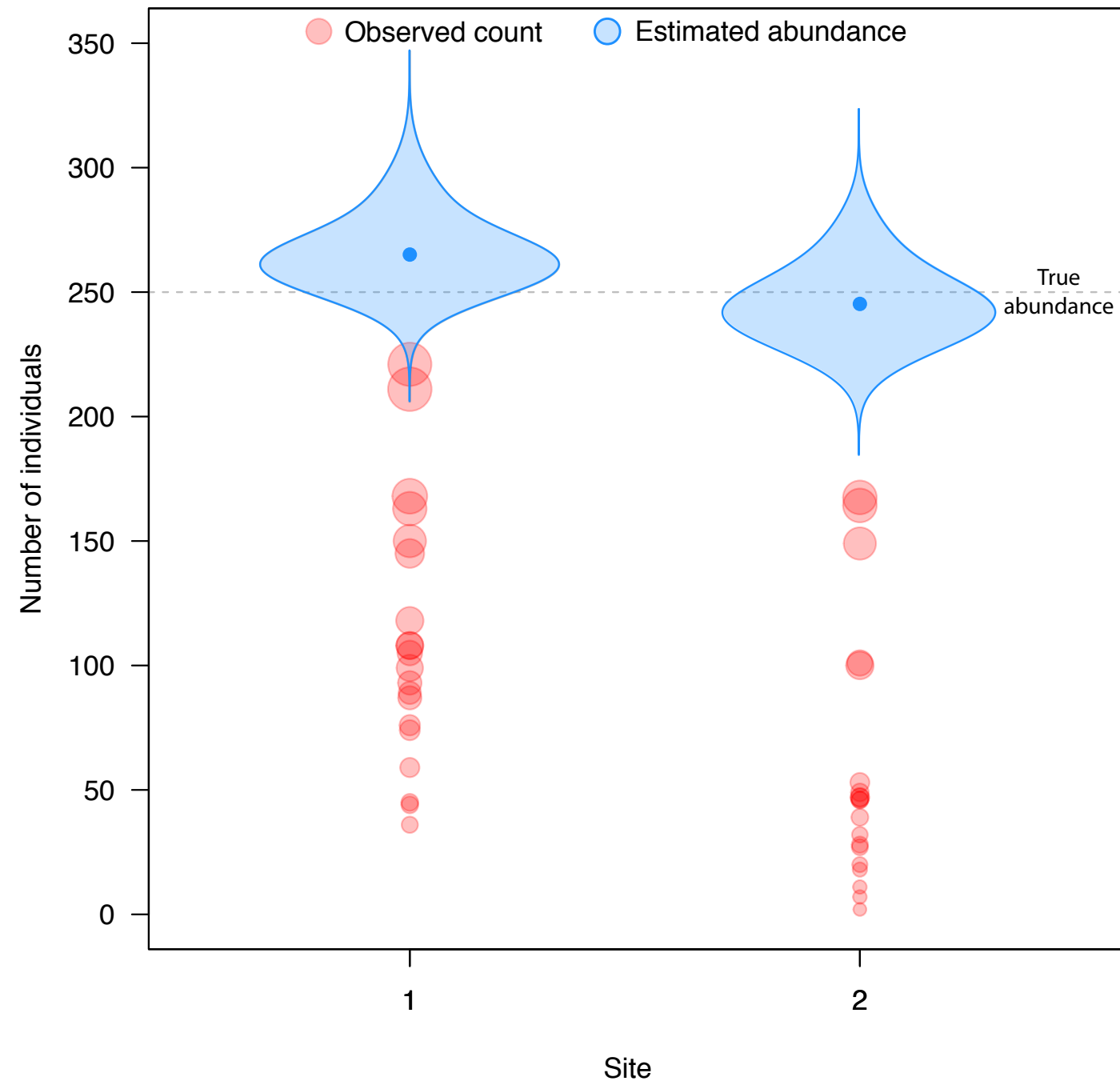
N — latent variable, total abundance

p_i — detection probability for survey i

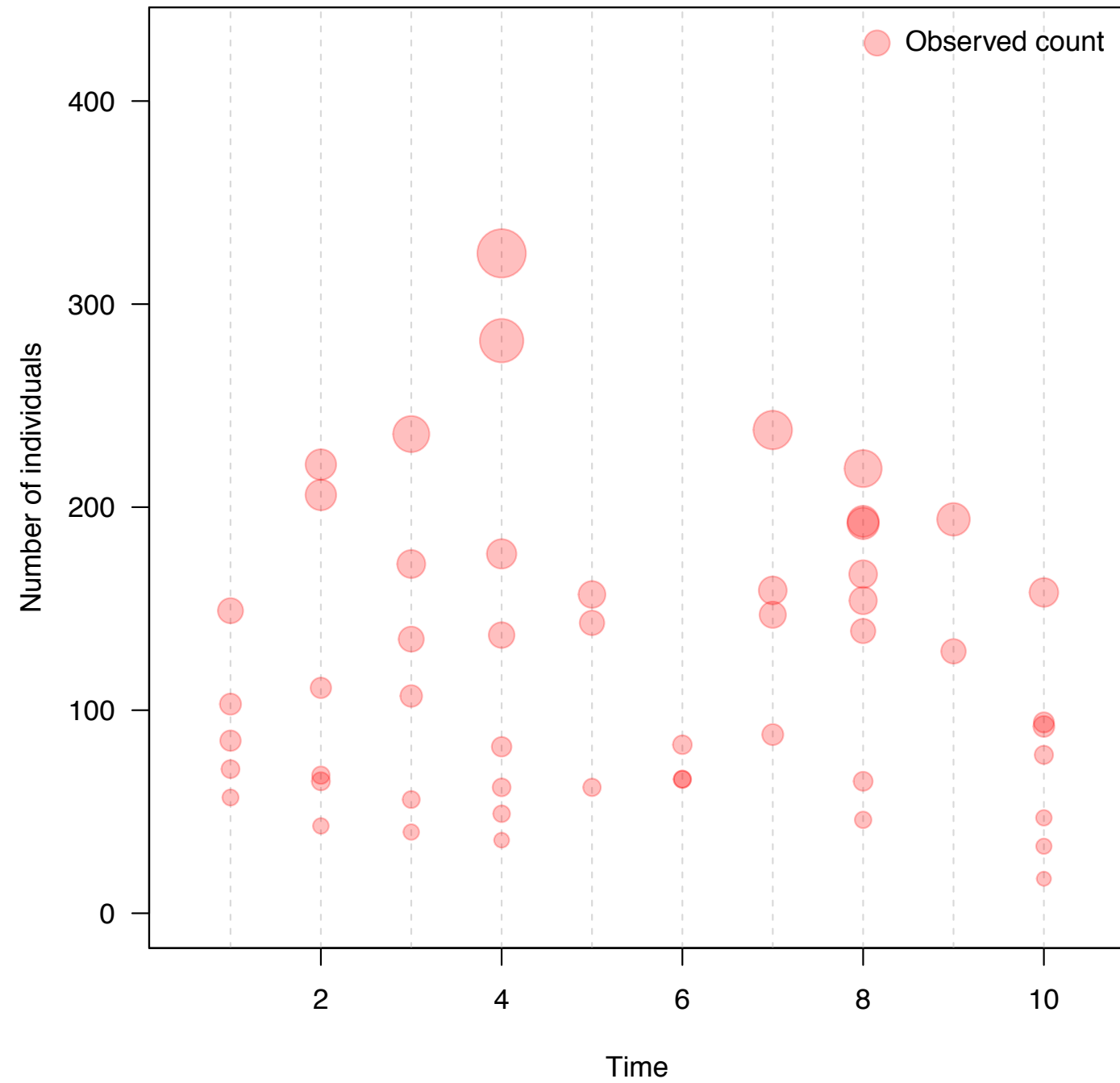
$\mathbf{W}$ — design matrix for detection probability

$\boldsymbol{\alpha}$ — coefficients for detection probability

$\mathcal{N}$ -mixture Model



Population Dynamics



Dynamic $\mathcal{N}$ -mixture Model

Model N_t as a function of N_{t-1} :

$$y_{it} \sim \text{Binom}(N_t, p_{it})$$

Observation model

$$N_t \sim \text{Pois}(\theta_t)$$

Biological process model

$$\theta_t = \lambda N_{t-1}$$

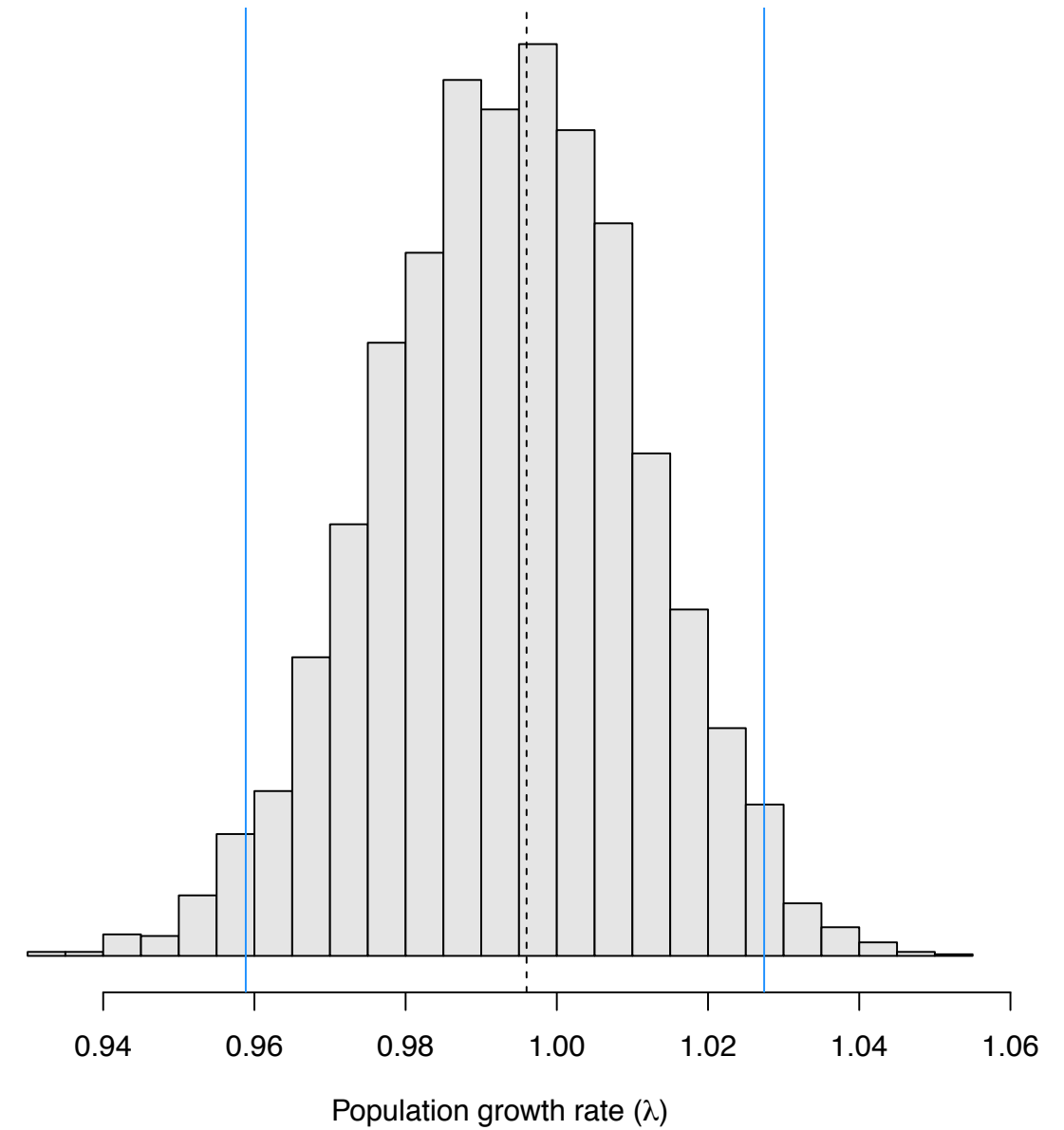
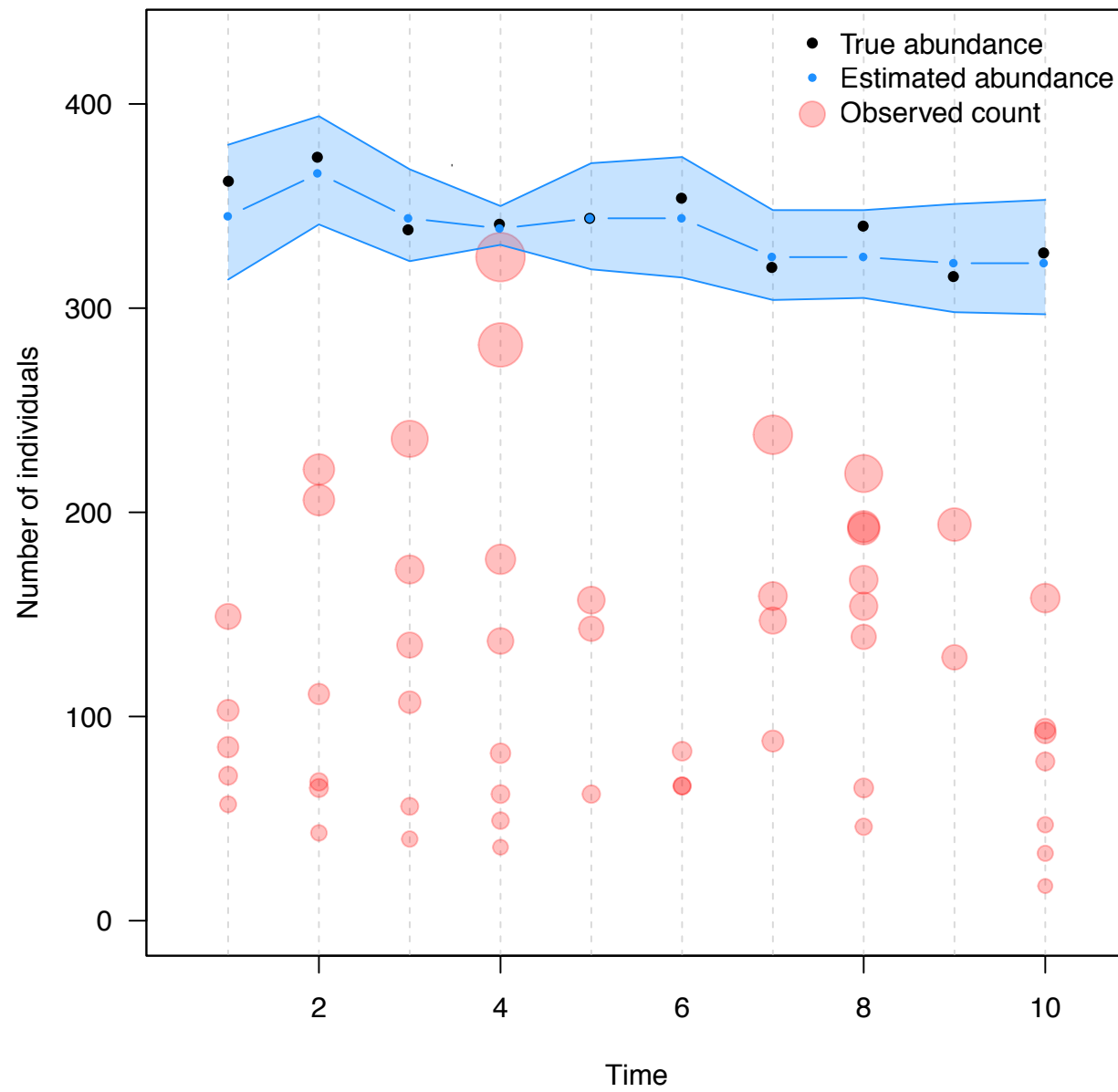
Same as $N_t = \lambda N_{t-1}$

$$\text{logit}(\mathbf{p}) = \mathbf{W}\boldsymbol{\alpha} + \boldsymbol{\epsilon}$$

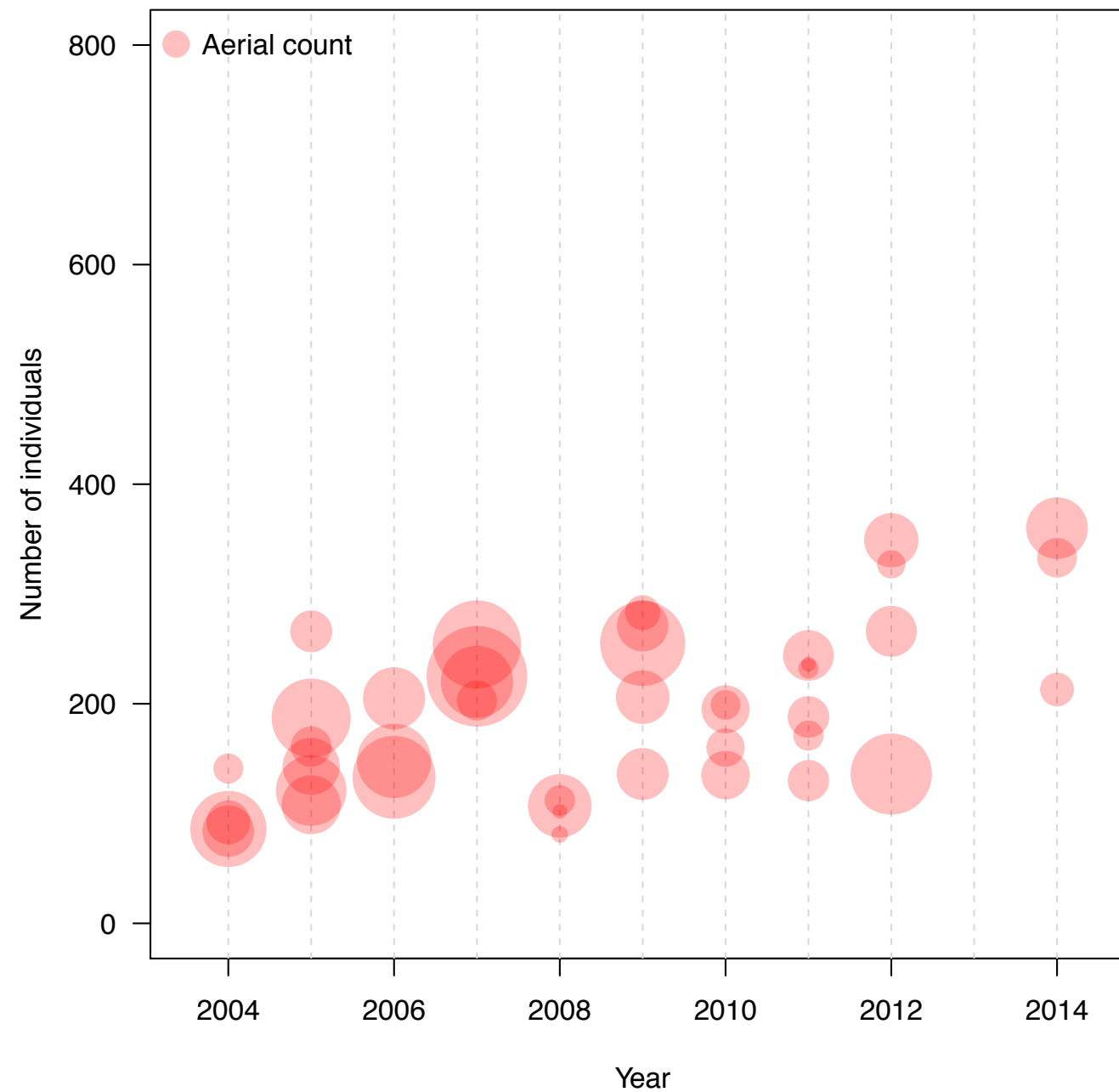
Detection model

(Plus other stuff...)

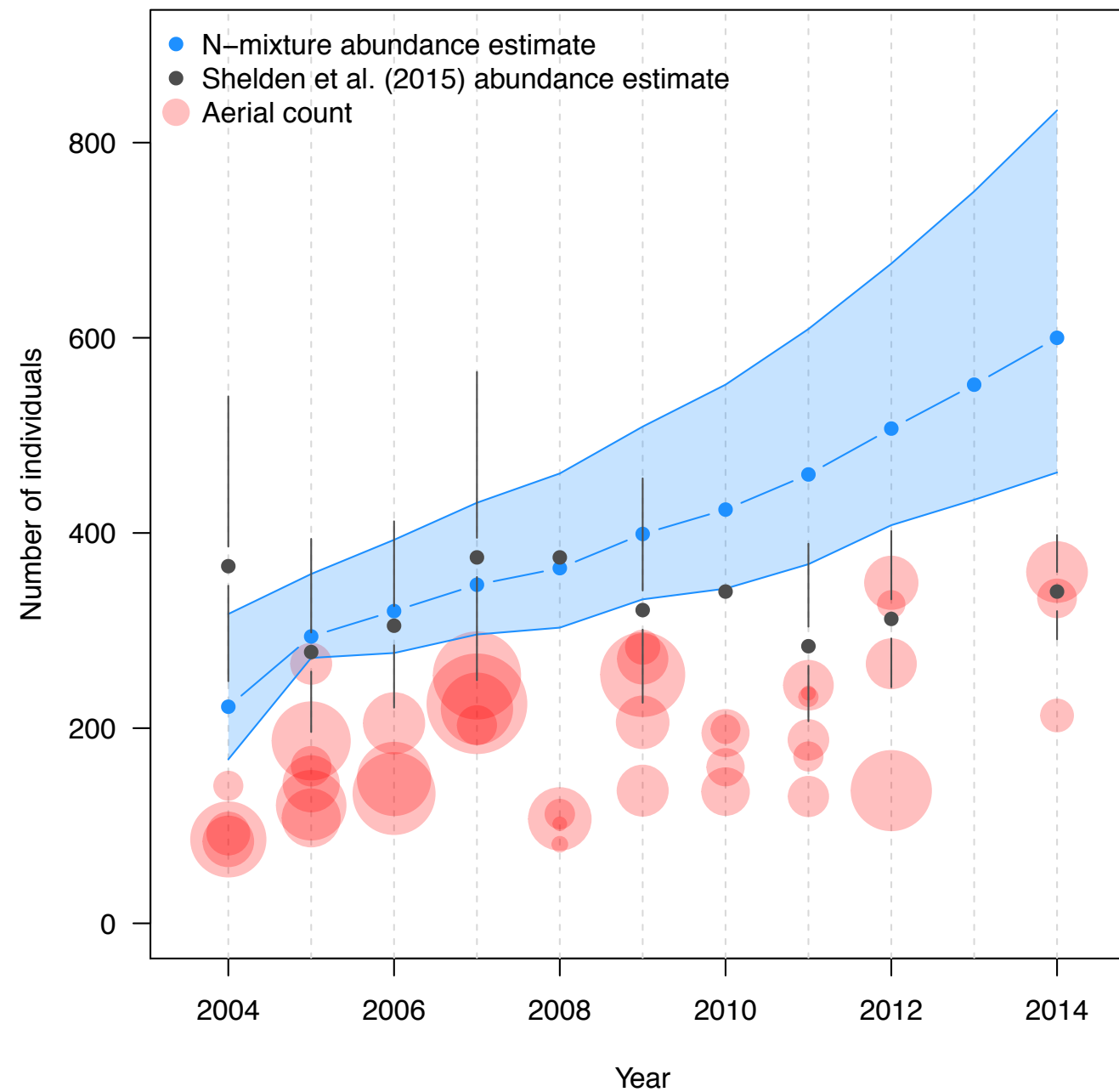
Dynamic $\mathcal{N}$ -mixture Model



Cook Inlet Beluga Population Dynamics



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Assumptions

- Distributional forms and functional relationships
- Population closure
- No false positives



QUESTIONS?